Justice-Chain SOS device

Raspberry Pi shopping lists: 3A+, 3B+ and Pi 4

14 October 2026 · Replaces the Pi Zero 2 WH, which is not available

Any of these three boards works with the existing implementation plan without changes. They all have the 40-pin header already attached (no soldering), use the same camera software (picamera2) and the same hardware video encoder as the Pi Zero 2, and keep the same wiring: the SOS button on pin 11 and the status LED on pin 13. Pick one board, then buy only that board's list.

All prices are estimates in ₹. Check them on Robu.in or Robocraze before ordering.

Choosing the board

	Pi 3A+	Pi 3B+	Pi 4 (2 GB)
RAM	512 MB (enough)	1 GB	2 GB
Size	65 × 56 mm (smallest)	85 × 56 mm	85 × 56 mm
USB ports	1 × USB-A	4 × USB-A	2 × USB 3 + 2 × USB 2
Power input	micro-USB, 5V 2.5A	micro-USB, 5V 2.5A	**USB-C, 5V 3A**
Rough battery life, 10,000 mAh bank (to be measured)	~15–25 h idle	~12–15 h	~8–10 h
Heat	Low	Medium	Hot, needs heatsink + fan
Board price (₹)	2,800–3,800	3,500–4,800	4,500–5,800
Full build (₹)	4,000–9,500	4,500–10,500	6,000–12,500
Good for	Closest to the Zero: small, low power	Middle choice	Fastest to develop on, easiest to find

Buying order if stock is limited: Pi 3A+ → Pi 4 (2 GB) → Pi 3B+.

Do not buy

- **Raspberry Pi 5**: no hardware video encoder, and it uses a different (22-pin) camera connector.
- Any camera marked "22-pin", "for Pi 5" or "for Pi Zero": wrong cable ends for these boards.
- Raspberry Pi HQ camera or Arducam "Pivariety" / 16 MP / 64 MP models: need extra lenses or drivers.

Option 1: Raspberry Pi 3A+

Smallest, best battery life

#	Item	Qty	Notes	Approx. ₹
1	Raspberry Pi 3A+	1	40-pin header already attached	2,800–3,800
2	**Camera Module 3 Wide** (best) **or** OV5647 160° wide (budget)	1	Must say "15-pin standard CSI". The cable comes with it	3,000–4,500 / 900–1,800
3	Mini USB microphone (dongle)	1	Uses the board's only USB port	200–500
4	microSD card, 32 GB, Class 10 / A1 (e.g. SanDisk Ultra)	2	One spare	800–1,200
5	microSD card reader	1	Only if your laptop has no slot	150–300
6	micro-USB cable, short and thick	1	For power	100–200
7	Power bank, 10,000 mAh, **5V 2.4A output**	1	For portable use	800–1,500
8	Official 5V 2.5A micro-USB power adapter	1	For desk work. Weak chargers cause crashes	500–800
9	Heatsink set (stick-on)	1		50–150
10	Tactile push button, 12 × 12 mm	3	SOS button + spares	30–60
11	5 mm LEDs (red/green)	4	Status light	20–40
12	330 Ω resistors	5	One per LED	10–20
13	Mini breadboard	1		60–120
14	Jumper wires, female-to-male + male-to-male	1 set each	Female ends fit the Pi pins	80–150
15	Small project box or case	1	Holes for lens, mic, button, LED and cable	200–600
16	*Optional:* 300 mm camera cable	1	If the camera sits away from the board	100–200

Estimated total: about ₹6,000–9,500 with Camera Module 3 Wide, or ₹4,000–6,500 with the OV5647.

Notes

- Only one USB port: it is used by the microphone. Setup is done over SSH, so no keyboard is needed.
- 512 MB RAM is enough for recording, locking and uploading 45-second clips.
- Check the camera listing says **15-pin standard Raspberry Pi CSI**.
- Wiring: SOS button between pin 11 and a ground pin; LED + 330 Ω resistor on pin 13.

Option 2: Raspberry Pi 3B+

Balanced

#	Item	Qty	Notes	Approx. ₹
1	Raspberry Pi 3B+	1	40-pin header already attached	3,500–4,800
2	**Camera Module 3 Wide** (best) **or** OV5647 160° wide (budget)	1	Must say "15-pin standard CSI". The cable comes with it	3,000–4,500 / 900–1,800
3	Mini USB microphone (dongle)	1		200–500
4	microSD card, 32 GB, Class 10 / A1 (e.g. SanDisk Ultra)	2	One spare	800–1,200
5	microSD card reader	1	Only if your laptop has no slot	150–300
6	micro-USB cable, short and thick	1	For power	100–200
7	Power bank, 10,000 mAh, **5V 2.4A output**	1	For portable use	800–1,500
8	Official 5V 2.5A micro-USB power adapter	1	For desk work	500–800
9	Heatsink set (stick-on)	1		50–150
10	Tactile push button, 12 × 12 mm	3	SOS button + spares	30–60
11	5 mm LEDs (red/green)	4	Status light	20–40
12	330 Ω resistors	5	One per LED	10–20
13	Mini breadboard	1		60–120
14	Jumper wires, female-to-male + male-to-male	1 set each	Female ends fit the Pi pins	80–150
15	Case for Pi 3B+ (with camera-cable slot) or a project box	1		200–600
16	*Optional:* 300 mm camera cable	1		100–200

Estimated total: about ₹6,500–10,500 with Camera Module 3 Wide, or ₹4,500–7,500 with the OV5647.

Notes

- Four USB ports, so a keyboard can be plugged in alongside the mic if needed.
- Check the camera listing says **15-pin standard Raspberry Pi CSI**.
- Wiring: SOS button between pin 11 and a ground pin; LED + 330 Ω resistor on pin 13.

Option 3: Raspberry Pi 4 Model B (2 GB)

Easiest to find, fastest to develop on

#	Item	Qty	Notes	Approx. ₹
1	Raspberry Pi 4 Model B, **2 GB**	1	4 GB also works, but costs more for no gain	4,500–5,800
2	**Camera Module 3 Wide** (best) **or** OV5647 160° wide (budget)	1	Must say "15-pin standard CSI". The cable comes with it	3,000–4,500 / 900–1,800
3	Mini USB microphone (dongle)	1		200–500
4	microSD card, 32 GB, Class 10 / A1 (e.g. SanDisk Ultra)	2	One spare	800–1,200
5	microSD card reader	1	Only if your laptop has no slot	150–300
6	**USB-C** cable	1	For power	150–300
7	Power bank, 10,000 mAh, **USB-C 5V 3A output** (PD type)	1	For portable use. The Pi 4 needs more power than the Pi 3 boards	1,200–2,000
8	Official Raspberry Pi 15W USB-C power supply	1	For desk work	700–1,000
9	**Heatsink + small 5V fan**, or a case with a fan	1	The Pi 4 runs hot while recording	300–600
10	Tactile push button, 12 × 12 mm	3	SOS button + spares	30–60
11	5 mm LEDs (red/green)	4	Status light	20–40
12	330 Ω resistors	5	One per LED	10–20
13	Mini breadboard	1		60–120
14	Jumper wires, female-to-male + male-to-male	1 set each	Female ends fit the Pi pins	80–150
15	Case for Pi 4 with fan mount and camera-cable slot, or a vented project box	1	Skip if the item above is already a fan case	300–700
16	*Optional:* 300 mm camera cable	1		100–200

Estimated total: about ₹8,000–12,500 with Camera Module 3 Wide, or ₹6,000–9,500 with the OV5647.

Notes

- Power is the most common Pi 4 problem: use a 5V 3A supply, otherwise it slows down or reboots.
- Check the camera listing says **15-pin standard Raspberry Pi CSI**.
- Wiring: SOS button between pin 11 and a ground pin; LED + 330 Ω resistor on pin 13.

Camera options

All of these work with Raspberry Pi OS (Bookworm) and picamera2 with no extra drivers. None of the ribbon cameras has a microphone, so add the mini USB mic from the list.

Camera	Sensor	Focus / view	Approx. ₹	Verdict
Camera Module 3 (official)	Sony IMX708, 12 MP	Autofocus, HDR, 75°	2,500–3,500	Best quality. Autofocus helps when the device moves
Camera Module 3 Wide	IMX708, 12 MP	Autofocus, **120°**	3,000–4,500	**Recommended.** Captures more of the scene
Camera Module 3 NoIR / NoIR Wide	IMX708, no IR filter	As above	3,000–4,500	Only useful with IR lights at night; odd colours by day
Camera Module 2 (official or clone)	Sony IMX219, 8 MP	Fixed focus, 62°	1,200–2,800	Good middle option
OV5647 "Rev 1.3"	OmniVision OV5647, 5 MP	Fixed focus, ~54°	400–900	Cheapest; fine for a demo
OV5647 wide / fisheye (e.g. Waveshare 160°)	OV5647, 5 MP	Fixed, 130–160°	900–1,800	**Recommended budget pick.** Cheap and wide
OV5647 night vision (IR-cut + IR LEDs)	OV5647, 5 MP	Fixed	1,000–1,800	Sees in the dark; IR LEDs glow faintly red and drain the battery
USB webcam with built-in mic (e.g. Logitech C270)	—	Fixed, 60°	1,500–2,000	Video and sound in one plug (no separate mic). Bulkier; recorder code uses ffmpeg instead of picamera2

Our pick

- **Camera Module 3 Wide** if the budget allows.
- Otherwise an **OV5647 160° wide** camera for a cheap build.
- Either way, add the mini USB microphone.

Before you order

- Board: 40-pin header already attached (true for every 3A+, 3B+ and Pi 4).
- Camera: listing says "15-pin" / standard Raspberry Pi CSI, not "22-pin" or "Pi 5".
- Power bank: 5V 2.4A output for the Pi 3 boards, USB-C 5V 3A output for the Pi 4.
- Delivery usually takes 3–7 days, which is the first bottleneck in the plan.